STAT 654
Bayesian Computation

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Traditional Inference

- Given data Y, and unknown parameter β, how to estimate β?
- We can find an unbiased estimator of β.
- Possible methods may be least squares (LS) or maximum likelihood estimates (MLE) by looking at the data likelihood.
- Review earlier class material for reference.

Introduction to Bayesian Inference

- Bayesian methods trace their origin to the *Bayes' Theorem*.
- The important quantities are:
- $p(y \mid \beta)$: Data Likelihood.
- $p(\beta)$: Prior.
- $p(\beta \mid y)$: Posterior.
- $p(y)$: Marginal Distribution.
- According to the Bayes' Theorem:
  $p(\beta \mid y) = \frac{p(\beta, y)}{p(y)} = \frac{p(y \mid \beta) p(\beta)}{p(y)} \propto p(y \mid \beta) p(\beta)$

Bayesian Inference

- Let $Y_1, ..., Y_n \overset{\text{i.i.d.}}{\sim}$ Bernoulli (β).
- Then the data likelihood is: $\prod_{i=1}^{n} P(Y_i = y_i \mid \beta)$
$= \prod_{i=1}^{n} \beta^{y_i}(1-\beta)^{1-y_i} = \beta^{\sum y_i}(1-\beta)^{n-\sum y_i}$
- Before taking data, one has beliefs about the value of the parameters and one models beliefs in terms of a prior distribution.
- We assume a functional form for this prior.
- After data have been observed, we update our beliefs about the parameter by computing the posterior distribution.
- One summarizes this probability distribution to perform inferences.
- Also one may be interested in predicting the likely outcomes of a new sample taken from the population.

Bayesian Inference

- Let $a, b > 0$, and let $\beta \sim Beta(a, b)$.
- So, $p(\beta) = \frac{1}{B(a,b)}\beta^{a-1}(1-\beta)^{b-1}, \;\; 0 < \beta < 1$.
- Note that $B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$
- Thus, $p(\beta) \propto \beta^{a-1}(1-\beta)^{b-1}, \;\; 0 < \beta < 1$.
- The posterior: $p(\beta \mid y) \propto Beta(\beta \mid a + \sum y_i, b + n - \sum y_i)$.

Learning About the Proportion of Heavy Sleepers

- Interested in learning about the sleeping habits of American college students.
- Let p represent the proportion of this population who sleep at least eight hours.
- We are interested in learning about the location of p.
- Suppose that our prior density for p is denoted by $g(p)$.
- If we regard a "success" as sleeping at least 8 hours and we take a random sample with s successes and f failures, then the likelihood function is: $L(p) = p^s(1-p)^f, 0 < p < 1$.
- The posterior density for p, by Bayes' Rule, is obtained, up to a proportionality constant, by multiplying the prior density by the likelihood: $g(p \mid data) \propto g(p)L(p)$.

Beta Prior

- Since the proportion is continuous in the interval $(0, 1)$, we can assume a $Beta(a, b)$ for p.
- So, prior for p: $g(p) \propto p^{a-1}(1-p)^{b-1},\ 0 < p < 1.$
- We can ignore the normalizing constant.
- Combining this beta prior with the likelihood function, one can show that the posterior density is also of the beta form with updated parameters $a + s$ and $b + f$:
  $g(p \mid data) \propto p^{s+a-1}(1-p)^{f+b-1},\ 0 < p < 1.$
- This is an example of a *conjugate analysis* where the prior and posterior densities have the same functional form.

Prediction

- Suppose we are interested in predicting the number of heavy sleepers $\tilde{y}$ in a future sample of $m = 20$ students.
- If the current beliefs about p are contained in the density $g(p)$, then the predictive density of $\tilde{y}$ is given by $f(\tilde{y}) = \int f(\tilde{y} \mid p) g(p) dp$
- If g is a prior density, then we refer to this as the *prior predictive density*.
- If g is a posterior density, then f is a *posterior predictive density*.

Predictive Density

- Say p has a $Beta(a, b)$ prior.
- We can analytically integrate out p to get a closed-form expression for the predictive density:
  $f(\tilde{y}) = \int f_B(\tilde{y} \mid m, p) g(p) dp$
  $= \begin{pmatrix} m \\ \tilde{y} \end{pmatrix} \frac{B(a+\tilde{y}, b+m-\tilde{y})}{B(a,b)}, \quad \tilde{y} = 0, ..., m.$
  where $B(a, b)$ is the Beta function.

The Normal-Normal Model

- The normal distribution $\mathcal{N}(\mu, \sigma^2)$ has p.d.f. $\frac{1}{\sqrt{2\pi\sigma^2}} exp(-\frac{1}{2\sigma^2}(x-\mu)^2)$ for $x \in \mathscr{R}$.
- Assume $X_1, ..., X_n \mid \mu \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$.
- Also, assume $\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$.
- This setup is referred to as the Normal-Normal model.
- It turns out that the posterior $p(\mu \mid \boldsymbol{x}) \sim \mathcal{N}(\mu_p, \sigma_p^2)$.
- Here $\mu_p = \frac{\frac{\sum_{i=1}^n x_i}{\sigma^2} + \frac{\mu_0}{\sigma_0^2}}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}}$, and $\sigma_p^2 = \frac{1}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}}$.
- Thus, the normal distribution is, itself, a **conjugate prior** for the mean of a normal distribution with known variance.

What is Markov Chain Monte Carlo

- Markov Chain - where we go next only depends on our last state (the Markov property).
- Monte Carlo - algorithms that rely on repeated random sampling to obtain results.

Gibbs Sampling

- Let $p(\theta_1, \theta_2)$ be a p.d.f. or a p.m.f. that is difficult to sample from directly.
- Suppose that we can easily sample from the conditional distributions $p_1(\theta_1|\theta_2)$ and $p_2(\theta_2|\theta_1)$.
- The *Gibbs sampler* proceeds as follows:
- 1. set θ_1 and θ_2 to some initial starting values.
- 2. sample $\theta_1|\theta_2$, then sample $\theta_2|\theta_1$, then $\theta_1|\theta_2$, and so on.
- Intuition: Repeated sampling from the distributions, namely $p_1(\theta_1|\theta_2)$ and $p_2(\theta_2|\theta_1)$, will generate a sequence of θ_1 and θ_2 values, say $(\theta_1^{(1)}, \theta_2^{(1)}), \ldots, (\theta_1^{(L)}, \theta_2^{(L)})$.
- Let us throw away the first B samples (burn-in), so we are left with $(L - B)$ samples (post burn-in).
- These post burn-in samples can be shown to be approximate samples from the joint distribution, i.e., $p(\theta_1, \theta_2)$.

Gibbs Sampling Example

- Suppose $Y \sim \mathcal{N}(\mu, \sigma^2)$, and we obtain sample $y_1, ..., y_n$.
- The priors are $f(\mu, \sigma^2) = f(\mu) \times f(\sigma^2)$, with $f(\mu) \propto 1$ and $f(\sigma^2) \propto 1/\sigma^2$.
- Based on a sample, we want to obtain the posterior distributions of μ and σ^2 using the Gibbs sampler.
- The *posterior conditionals* are the following:
  $\mu \,|\, \sigma^2, y_1, ..., y_n \sim \mathcal{N}(\bar{y}, \frac{\sigma^2}{n})$
  $\sigma^2 \,|\, \mu, y_1, ..., y_n \sim IG(\frac{n}{2}, \frac{\sum_{i=1}^{n}(y_i - \mu)^2}{2})$, where $IG(a, b)$ denotes the Inverse-Gamma distribution with parameters a and b.